

Telecom Customer Churn Analysis & Predictive Modeling

End-to-End Analytics, Iterative Machine Learning Pipeline, & Business Intelligence Report

1. Executive Summary

Customer churn represents a critical revenue risk in the telecommunications industry. This project combines exploratory data analysis, an iterative machine learning pipeline (Decision Tree to Optimized Random Forest), and an interactive Power BI dashboard suite to uncover historical churn drivers and predict high-risk active accounts for proactive retention.

2. Business Problem & Objectives

To protect recurring revenue and maintain customer lifetime value, leadership established core analytical questions that this project addresses:

- **Why are customers leaving?** (Diagnosing root causes such as competitor offerings, pricing tiers, and support quality).
- **Which customers are likely to churn?** (Predicting active accounts at risk via machine learning risk probabilities).
- **How much revenue is at risk?** (Quantifying potential financial impact on recurring revenue streams).
- **Which customer segments require immediate attention?** (Isolating high-risk cohorts like month-to-month and fiber optic users).
- **What actions should the retention team take?** (Providing targeted, actionable intervention workflows).

3. Key Performance & Financial Metrics

Historical Portfolio Snapshot (Customer Churn Overview)

7,043 TOTAL CUSTOMERS	1,869 (26.5%) TOTAL CHURN RATE	\$3.68M HISTORICAL CHURN REV
73.46% RETENTION RATE	32.39 Mo AVG CUSTOMER TENURE	\$3,034.38 ARPU

Predictive Risk Snapshot (Churn Analysis - Predictions)

5.17K ACTIVE CUSTOMERS	149 POTENTIAL CHURNERS	\$154.56K POTENTIAL REVENUE RISK
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4. Machine Learning Pipeline & Model Iteration

To ensure high predictive accuracy and robustness, the machine learning workflow followed an iterative optimization process:

- **Baseline Modeling (Decision Tree & Random Forest):** Initial experimentation began with a standard Decision Tree classifier to map primary decision splits, followed by a baseline Random Forest model to minimize overfitting and improve stability across classes.
- **Hyperparameter Tuning (GridSearchCV):** The baseline Random Forest model was optimized using Grid Search cross-validation to tune critical parameters (such as number of estimators, max depth, and minimum samples split), yielding the top-performing base configuration.
- **Feature Selection & Final Optimization:** Feature importance metrics were extracted from the best-performing model to isolate core predictors (e.g., contract type, monthly charges, tenure, and payment method). A final optimized model was retrained exclusively on these **best features**, reducing noise and complexity while stabilizing classification metrics (~83% accuracy).

5. Key Findings & Analytical Insights

A. Historical Drivers

- **Primary Churn Categories:** Competitor actions lead departures with 841 churns, followed by Dissatisfaction (321) and Attitude issues (314).
- **Specific Complaints:** Customers frequently cited competitors offering better devices (313) or better overall offers (311), alongside poor customer support experiences.
- **Tenure & Pricing Peak:** Churn rates peak significantly during the first year of subscription (1-12 months) and among accounts with high monthly charges (\$51-\$100 range).

B. Predictive Insights

- **Contract Structure Risk:** 100% of high-risk active customers are bound to Month-to-Month contracts, proving contract flexibility is the primary structural risk factor.
- **Payment & Service Concentration:** Customers utilizing Bank Withdrawal represent 88.5% (128 of 149) of high-risk predictions, while Fiber Optic users comprise 88.59% (132 users) of the pool.

6. Recommended Actions for Retention Teams

- **Targeted Month-to-Month Conversions:** Immediately deploy promotional incentives to transition high-risk month-to-month subscribers onto 1-year or 2-year locked agreements.
- **Fiber Optic Satisfaction Audits:** Launch technical audits and proactive check-ins for fiber subscribers to resolve connection reliability and pricing complaints before cancellation.
- **Billing Friction Review:** Investigate payment process friction associated with Bank Withdrawal accounts, as this billing channel strongly correlates with high churn probability models.
- **Operationalized Call Lists:** Hand off the exported Power BI high-risk tracking table directly to retention call centers for immediate, high-priority customer outreach.